Design and Development of IoT based Robotic Arm by using Arduino

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Abstract: In the current scenario machines and robots are playing an important role in the automation industry. This paper is presenting the process through which robotic arm is made with the help of Arduino and Potentiometer for controlling and coordinating the industrial processes. Here, we realize that the robotic arm has the ability to move in four directions with the help of servo motors. The movement of servo motor, UNO is responsible which converts the analog signal into the digital signal which is further received by servo motor. This project discuss about the technical imputation, the issue related with the implications and application of robotic arm in the field of automation of industries. It can be used as a artificial arm for a human who have lost his hand in any accident.

Keywords: IoT (Internet of Things), Robotic Arm, Potentiometer, Servo Motor, Arduino, MEMS (Micro-Electro-Mechanical Systems)

I. Introduction

Nowadays, everything is somehow related to technology, every people are totally depends on technology and automation. In today's world, human work is increasing day by day and become tough to handle the lot of task in less time with high accuracy. This is the reason, Robotics comes into usage which reduces the human effort, it works with high precision and give valuable output, also helps to save the time, money and resources. Robots improve the quality and efficiency of task and leave the humans for other innovative task. They can work 24 hours in a day and can do a single work repeatedly without a single complain and without getting tired.[11]

Robotics is the combined work of mechanical engineering, electrical engineering and computer science. Robot is a machine designed to perform a particular job, coded with some programming language by the user. Here, Arduino language is used which is based on simple processing and it is same as C programming language. The code is written into Arduino IDE then uploaded on the Arduino board in which small tiny integrated circuit called microcontroller is present to run the code. Robotic arm can be control by potentiometer through which left, right, up and down action can be done with four degree of freedom. Robotic arm consists of four servo motor which helps to move the arm.

There can be many different types of robotic arm for different purposes. Robotic arm is a copy of human arm which can do rotational motion and translation motion as human arm can do. Nowadays, Robotic arm are the major part of industries to pick and place the things from one place to another. They reduce the danger and difficulties for humans and they can be used in hazardous place where humans cannot go or not safe for the humans.

This project is based on IoT. IoT[13] is a trending technology nowadays. Many multinational companies are working on it and providing services to the customers. Many hotels, apartments and residencies etc. are fully automated with the help of IoT.[9][21] It is used in robotic arm to collect and exchange or transfer the data according to which robotic arm can take action.[8][10]

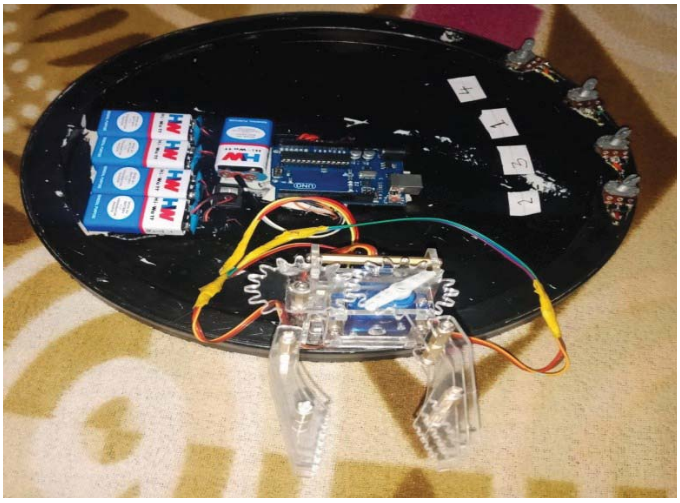

Figure 1: Complete Structure of Robotic Arm

2. Proposed Work

Robotic arm is a copy of human arm which can do rotational motion and translation motion as human arm can do. This block diagram is representing all the component connecting with each other. Here, four servo motor is connected with Arduino board, which is used for translation and rotational motion and power source (Battery) and other four potentiometers is also connected with Arduino board.[1][25]

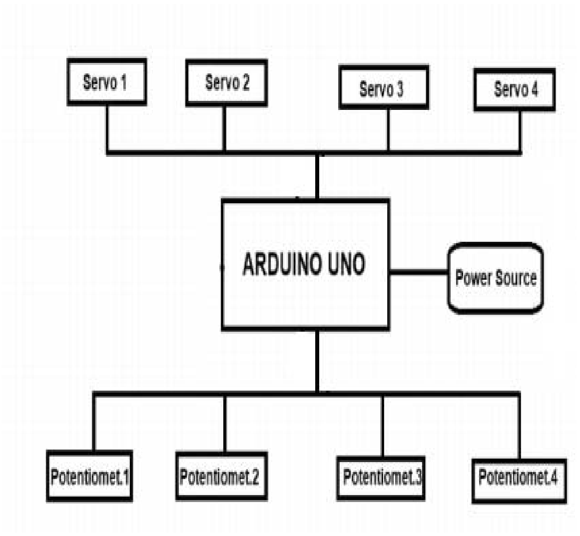

Figure 2: Block Diagram of Robotic Arm

3. Robotic Arm

A robotic arm is a type of mechanical arm that performs the same function as a human arm. It is very helpful and reduces human efforts. It is very useful for complex and dangerous tasks such as welding (tungsten welding), gripping, spinning, etc. Let us take an example of cleaning the toilet seat and waste management where we train the robot's hand to do the work unknowingly that was once done by a human. In this way, it is also ensuring cleanliness. This is well needed in work where accuracy, consistency, repetition is required. [7][12]

3.1 Types Of Robotic Arm

1. **Cartesian Robot:** Used to operate and work with machine tools and arcs.
2. **Cylindrical Robot:** It is mostly used for assembly purpose operation, machine tool handling, spot welding. This is a robot in which the axes are like a cylindrical frame of reference.
3. **Spherical Robots:** It is mainly used for Machine Tools, Spot Welding, Fettling Machines, Gas welding, and arc welding.
4. **Articulated robots:** Articulated robots can range from simple two-joint structures to 10 or more interactive pair systems. They are managed by the expansion of means, including electric motors.
5. **Parallel Robot:** A parallel manipulator is a mechanism that uses multiple computer-controlled serial chains to support a platform or end effector. Probably, the simplest parallel manipulator is made from six linear actuators. The engineers who first designed and used the device were named Stewart Platform or Goff-Stewart Platform.
6. **SCARA Robot:** SCARA stands for Selective Compliance Assembly Robot Arm. Its arm is firmly positioned within the Z-axis and is flexible on the XY-axis, allowing it to adapt to XY-holes inside the axes. [17][18]

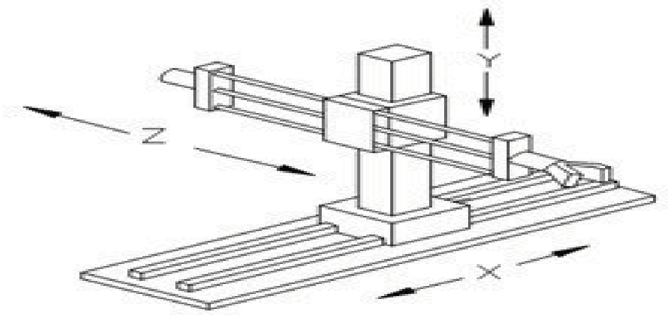

Figure 3: Cartesian Robotic Arm

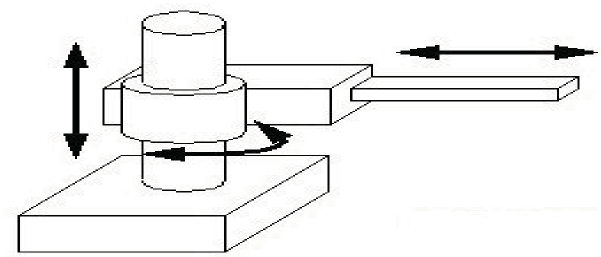

Figure 4: Cylindrical Robotic Arm

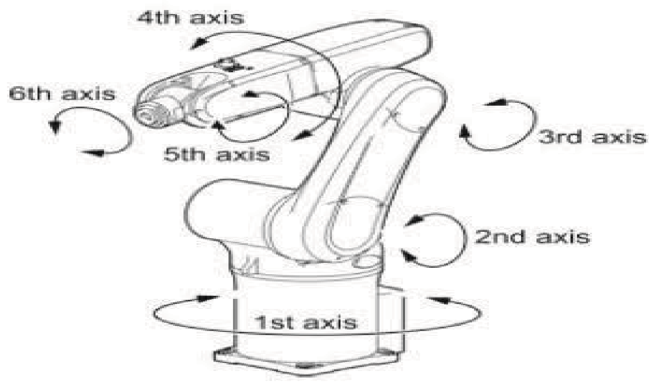

Figure 5: Articulated Robotic Arm

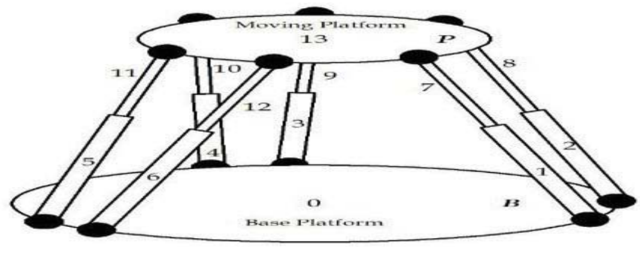

Figure 6: Parallel Robotic Arm

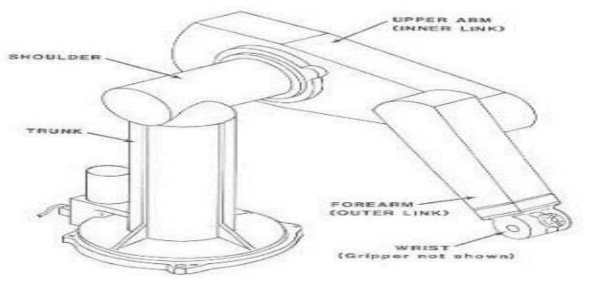

Figure 7: SCARA Robotic Arm

3.2 Evolution Of Robotic Arm

3.2.1Traditional Robots and Robotic Arms:

Now, the arrival of advance and automatic technology and the replacement of vacuum tubes with semiconductor transistors, the advancement of micro-circuits and more rapid computer structure, the platform for early new robot arm development has developed. The basic "position control mechanism" was designed in 1938. It was based on spray finishing robotic arm with five degree of freedom and was controlled by an electronic control system. Although the arm was not built by Pollard, his idea and interest towards the mechanical operation for automatic robotic arm rely on the ingeniousness of others. Both hands are useful and work-oriented on the joints in a way that is unique to their time and every movement.[12]

3.2.2 Multi-functional Robotic Arm:

The main motive of this robotic arm is to present the following works: Color classification, weightlifting, and Chinese script. And the motive of this robotic arm was to reduce costs as well as increase efficiency and performance of system. This robotic arm performs many task like a human hand. The ease of the design demonstrates almost a person's hand with its speed and speed improvement values. [6]

3.2.3 Development of robot-arm controllers using Hand Gestures Recognition:

In the interaction of humans with the machine, this robot arm controller uses image processing techniques. There are two methods to control the robot's hand, most often the desire to use hand gestures without the tools to help the system easily collect data (eg: gloves). The first method is to compare whole data collected in the database using a template matching algorithm, and second is the signature signal method, the distance between the arm edge and the arm's center. These essentially use gyro-sensors because the calculation of changes in angular values is obtained during the operation. Further operations will be determined based on these changes.[14][20]

3.2.4 Robot arm design and implementation based on Tactile Feedback Technology:

This includes the design of the tactile robot arm, which is prepared to hold and place objects. In this paper, the robotic arm is designed with four degrees of movement and can take objects or things with a certain load and place them in the right position. For ease of lifting objects, servomotors with 11 kg of torque are used. Programming is accomplished using Arduino programming on the ATMEGA-328 microcontroller.
The input is used remotely, one hand, made of polycarbonate, a potentiometer with a rotating fixed angle.
Potentiometer detects the degree of rotation and send the signals to the embedded controller or microcontroller correspondingly. [24]

3.2.5 Image advancement and identification system for robot arm control:

Image advancement and identification systems rely on sensors to continuously monitor the environment and respond accordingly. These are considered intelligent because their behavior simply triggers changes in the ecosystem and associated program actions. [16][19]

3.2.6 Six Degree of freedom Robotic arm for the intelligent robot:

Movement is dictated by a program that directs the proper motion and actions of a controller or robot that operates manually. How fast the robot unit accelerates or gains speed can contribute to operational improvement. Every added or updated movement implies ease of operation in some operating situation. Therefore, the idea that a circular rotating base can prove to be very useful because it is not as much expensive and has less time taking to rotate than to turn the assembly. This should be noted that this arrangement does not preclude other DOFs for early development. Each offer added or updated indicates flexibility. [3][4]

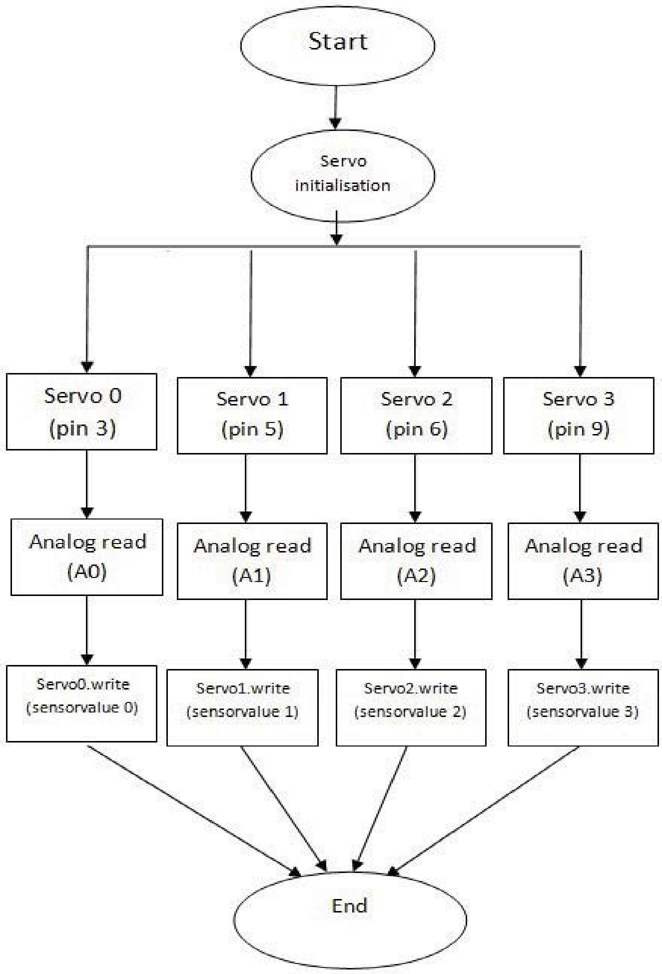

Figure 8: Flow Chart of Arduino Code

4. Advantages and Disadvantages

Advantage

- Keeping an eye and controlling of things become easy.
- Due to this precision, efficiency, speed and quality of product increases.
- Transfer of objects from one place to other become easy.
- Production of product increases.
- It is safer than human being during operation.
- Improve factories working condition.

Disadvantage

- Continuous power required.
- Lacking of security and privacy.
- Robotic arms are not creative and innovative.
- Experts need to handle.
- Fear of job losses increases.

5. Application

- May used for minor surgeries.
- Dislodge the hazardous objects without affecting humans.
- Picking and placing of objects from one place to another place.[15]
- May used for analysis and testing in lab.
- Can be used for polishing and welding.

7. Results and Discussion

The four servo are associated by the assistance of glue. These are associated by scaling and precise movement of Servo motor. With the help of piece of plastic a Servo is put in the center and stuck set up. The angle of movement is 0 to 180 degrees that provides the support of the arm. This support forms the primary connection on this servo the entire weight of other Servos were lifted. Earlier some issues came at first yet then it was fixed firmly with screws and made it. At that point piece of plastic is set on the above surface of the main Servo or more it a subsequent Servo is set over this piece and stuck this in the right position. The Servo turn simply be again straight angle (0 to 180 degree) motion.

This Servo structures the other connection of the hand subsequently shaping the respectable halfway point of automated arm. Every Servo is vital for other Servo connecting point shaping a hand with the goal that arm continued to be same steady and can utilize to pick and place the item effectively. The third Servo is fixed and gear firmly with the help of screws and glue. At that point fitting the sthird servo. At the end, the fourth or the last Servo is fixed at the surface of another piece of plastic. then we associate the third and fourth individually together and thus we set up and this servo structures the principle to pick and place gadget or items.

Now fit the jaw with the goal that we can easily pick the items with the help of jaw then every connections are stuck together and finally get a Robotic Arm prepared to associate with arduino pins and that can be constrained with the help of the 10k pot.[1][2][5]

Table 1: Quantitative analysis of the pick and place robot

Motor	M1	M2	M3	M4
Operation	Left and Right movement	Up and Down movement	Rotational movement	Arm and Gripper movement
Inertia	0.0121Kgm²	0.0091Kgm²	0.0022Kgm²	0.0045Kgm²
Mass	0.25kg	0.19kg	0.10kg	0.8kg
Radius	0.220m	0.220m	0.150m	0.075m
Mean error	5%	6%	5%	6%
Response time	1s	2s	1s	2s

The rotational inertia is calculated by:

$$\mathbf{RI = m \times r \times r}$$

Where m = Mass and r = Radius.

7. Conclusion

We succeed in this project successfully based on the Internet of Things.[22][23] The project should be observed and managed by using the Arduino. Robotic arm controller with a potentiometer we are giving four degrees of freedom and it will rotate 180 degrees. This project not only acknowledges the guidance sent but also accounts for the activity and can perform the task which can reduce the human efforts by performing the same task continuously and repeatedly.

8. Future Scope

- We can use IR sensors to automatically detect and avoid obstacles if robotic arm goes beyond line of sight.
- We can use MEMS technology for gesture controlled robotic arm.
- Potentiometer can be replaced by hand-gesture sensors.
- Robotic arm can be used as a artificial arm for a human who have lost his hand in any accident.
- Robotic arm can be used in medical field for surgeries and operations.
- Furthermore varieties and more flexibility to feature or replace any part consistent with the wants are often done to enhance its use and increase field of usage and to form it more universal or flexible.

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